TWO-WAY DIGITAL CROSSOVER FOR MULTI-DRIVER AUDIO

Per-Band Equalization and Comparative Evaluation of Complementary FIR and IIR Linkwitz–Riley (LR4) Crossover Designs

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Abstract —Many everyday audio products rely on more than one loudspeaker driver because no single transducer can reproduce the full audible range with the same efficiency, low distortion, and controlled dispersion. For this reason, modern systems split the audio spectrum into bands that can be routed to dedicated drivers (typically low-frequency and high-frequency units), and then tuned so that the final sound is balanced once the drivers are mounted in a real enclosure. This project presents a practical two-way digital crossover workflow for multi-driver audio, including a simple and adjustable per-band equalization stage that enables independent tuning of the low and high channels without redesigning the crossover. Two standard implementation families are considered: a linear-phase FIR solution and a low-latency IIR solution based on the Linkwitz–Riley approach. Using a consistent test pipeline, I verify correct band splitting, clean recombination, and the expected tuning behavior when per-band gains are changed. The results highlight an important practical point: while both approaches can deliver the intended frequency separation and band control, their phase and time-domain behavior can differ, which may affect transient reproduction and perceived clarity depending on the application constraints.

Keywords —Digital audio crossover; multi-driver loudspeakers; FIR filters; IIR filters; Linkwitz–Riley crossover; per-band equalization; phase and group delay; active loudspeaker DSP.

1. Introduction

Modern loudspeakers often rely on multiple drivers (e.g., woofer + tweeter) to cover the full audio band with acceptable distortion, directivity, and power handling. In these systems, a crossover is required to split a broadband input into frequency bands that match each driver's operating range, while still allowing the bands to recombine into a faithful overall response. In active loudspeakers, the crossover is implemented digitally and typically paired with per-band equalization (EQ) to compensate driver/enclosure characteristics and achieve the intended tonal balance.

A "good" digital crossover must satisfy two families of requirements. First, it must provide the intended spectral split: low-pass and high-pass branches should exhibit appropriate slopes and overlap around the crossover frequency, and the summed response should remain flat in magnitude. Second, it must preserve time-domain fidelity as much as the application requires: even if the magnitude of the sum is unity, phase and group delay behavior can differ substantially between implementations, with audible and system-level consequences (latency budget, transient reproduction, and alignment constraints).

This paper presents a compact and reproducible two-way DSP crossover block (crossover + per-band EQ) and compares two common implementation choices:

- Complementary FIR crossover, designed to achieve exact magnitude complementarity and (approximately) constant group delay consistent with linear-phase behavior, resulting in a summed response that corresponds to a pure delay.
- IIR Linkwitz–Riley 4th-order (LR4) crossover, widely used in practice for its efficiency and low latency, producing a unity-magnitude summed response but, in general, a frequency-dependent phase/group delay (all-pass behavior in the sum).

To make the differences clear and directly useful for design decisions, we evaluate both approaches using frequency-domain verification (branch magnitudes and summed magnitude; Fig. 1–3), time-domain reconstruction behavior (sum vs input; Fig. 4–5), and band-separated spectra and time signals on a representative audio clip, both before and after applying per-band EQ (Fig. 6–11). Finally, group-delay plots (Fig. 12–13) are used to interpret why "flat magnitude sum" does not necessarily imply "time-domain identity," and to link these behaviors to practical selection criteria.

The outcome is a set of plots and checks that can be applied immediately in daily crossover tuning work, and a clear discussion of the trade-offs between FIR and IIR crossovers when latency, computational cost, and phase behavior must be balanced.

2. Theoretical Background

2.1 Two-way digital crossover as a linear split–recombine system

A two-way crossover can be modeled as two linear time-invariant (LTI) branches [1] driven by the same input $x[n]$. The low- and high-band signals are

$$y_L[n] = (h_L * x)[n], \qquad y_H[n] = (h_H * x)[n]$$

and the reconstructed output is the sum

$$y_\Sigma[n] = y_L[n] + y_H[n] = \big((h_L + h_H) * x\big)[n].$$

In the frequency domain, this becomes

$$Y_\Sigma(e^{j\omega}) = \left(H_L(e^{j\omega}) + H_H(e^{j\omega})\right) X(e^{j\omega})$$

A crossover is typically designed so that the split is meaningful (each branch is dominant in its intended band) and the recombination remains controlled. In practice, "controlled recombination" can mean different targets depending on the application: a pure delay (linear phase), a unity-magnitude sum (all-pass), or a sum that is flat only over a limited frequency range.

2.2 Complementary FIR crossover and delay-perfect summation

In a complementary FIR design, the high-pass branch is often derived directly from the low-pass branch to enforce exact complementarity [1][2]. A common construction is

$$h_H[n] = \delta[n - D] - h_L[n]$$

which implies

$$H_H(e^{j\omega}) = e^{-j\omega D} - H_L(e^{j\omega}) \quad \Rightarrow \quad H_L(e^{j\omega}) + H_H(e^{j\omega}) = e^{-j\omega D}.$$

Therefore, the sum is a pure delay:

$$y_\Sigma[n] = x[n - D]$$

This property is stronger than a "flat magnitude sum": it guarantees that reconstruction differs from the input only by a constant time shift, which can be compensated exactly by aligning signals (e.g., removing an integer-sample delay). When the FIR branches are linear-phase, the branch delays are also approximately constant across frequency, which simplifies driver alignment and transient interpretation.

2.3 IIR Linkwitz–Riley 4th-order crossover and unity-magnitude summation

A widely used IIR choice in loudspeaker practice is the Linkwitz–Riley 4th-order (LR4) crossover [4], typically formed by cascading two 2nd-order Butterworth sections in each branch [3]. Conceptually,

$$H_{L,\mathrm{LR4}}(s) = \left(H_{L,\mathrm{BW2}}(s)\right)^2, \qquad H_{H,\mathrm{LR4}}(s) = \left(H_{H,\mathrm{BW2}}(s)\right)^2$$

with matched cutoff ω_c. LR crossovers are designed so that, around the crossover region, the two branches combine to a flat acoustic magnitude when summed (assuming coherent addition and proper polarity). In the discrete-time implementation, the same goal translates into an approximately unity-magnitude summed transfer function:

$$\left|H_L\left(e^{j\omega}\right) + H_H\left(e^{j\omega}\right)\right| \approx 1 \quad \text{over the band of interest}$$

Crucially, a unity-magnitude sum does not imply a pure delay. In general,

$$H_L\left(e^{j\omega}\right) + H_H\left(e^{j\omega}\right) = e^{-j\phi(\omega)}$$

i.e., the sum behaves as an all-pass with frequency-dependent phase $\phi(\omega)$. As a consequence, the recombined time-domain signal can differ from the original beyond any integer-sample alignment, even when magnitude errors are negligible. This distinction is central to interpreting reconstruction-error plots and group-delay behavior in the Results section.

2.4 Phase, group delay, and why stopband values are not meaningful

Phase and group delay provide time-domain insight for frequency-selective systems. The group delay of a transfer function $H\left(e^{j\omega}\right)$ is defined as

$$\tau_g(\omega) = -\frac{d}{d\omega}\arg\{H\left(e^{j\omega}\right)\}$$

and describes the local delay experienced by spectral components around ω. Linear-phase FIR filters exhibit approximately constant $\tau_g(\omega)$ in their passband, whereas IIR filters generally have frequency-dependent $\tau_g(\omega)$.

When analyzing branch group delays, values in the deep stopband must be treated carefully: where $\left|H\left(e^{j\omega}\right)\right|$ is extremely small, the phase is numerically unstable and physically irrelevant. For this reason, group-delay curves are meaningful primarily in the passband (or above a

practical magnitude threshold). In the paper figures, stopband regions are masked to avoid misleading spikes and to focus the interpretation on the frequency range that contributes to the output.

2.5 Per-band equalization in an active crossover chain

In an active loudspeaker DSP chain [4][5], each crossover branch is typically followed by a gain stage and optional equalization:

$$\widetilde{y_L}[n] = (g_L\, E_L)\{y_L[n]\}, \qquad \widetilde{y_H}[n] = (g_H\, E_H)\{y_H[n]\}$$

where g_L, g_H are band gains and E_L, E_H represent per-band EQ filters. The recombined signal becomes

$$\widetilde{y_\Sigma}[n] = \widetilde{y_L}[n] + \widetilde{y_H}[n]$$

Per-band EQ intentionally changes the spectral content of each branch and therefore the summed spectrum; it is not an "error" but part of real tuning practice. However, it imposes practical constraints: headroom must be preserved to avoid clipping, and the EQ should be interpreted together with the crossover slopes to ensure that the overall target response is achieved without undesirable overlap or excessive boost near the crossover.

3. Implementation and Reproducible Workflow

The project is implemented as a compact and reproducible MATLAB [6] pipeline that designs two crossover families (complementary FIR and IIR LR4), applies band-splitting with optional per-band equalization, computes objective verification metrics in both frequency and time domains, and exports publication-ready figures to the `figures` directory. All global parameters are centralized in a single configuration entry point (`cfg_default.m`) to prevent hidden settings and to ensure traceability between plots, audio exports, and design choices (sampling rate `cfg.Fs`, crossover frequency `cfg.fc`, FIR length `cfg.fir.N`, and EQ enablement/gains).

Software dependencies. The implementation relies on standard MATLAB plus Signal Processing Toolbox functions for resampling, spectral estimation, group-delay computation, and stable IIR filtering (e.g., `resample`, `pwelch`, `grpdelay`, `tf2sos`, `sosfilt`).

A key reproducibility constraint enforced by the pipeline is sampling-rate consistency. Since discrete-time filter coefficients depend explicitly on F_s, any mismatch between the input audio sampling rate and `cfg.Fs` invalidates both the crossover design and the subsequent comparisons. For this reason, the clip-processing stage includes an explicit sampling-rate check and, if required, resamples the input to `cfg.Fs` (implemented in `process_one_clip.m`). This prevents a common failure mode observed during development (e.g., processing a 44.1 kHz clip under a 48 kHz design) and guarantees that FIR and IIR results refer to the same discrete-time system.

Crossover design and numerical realization

Complementary FIR (delay-perfect summation by construction).

The FIR crossover is generated by first designing a linear-phase low-pass prototype using a windowed FIR method (Hamming-windowed `fir1`, configured by `cfg.fir.N` and $W_n = f_c/(F_s/2)$; see `design_fir_crossover.m`). The high-pass branch is then derived as the spectral complement of the low-pass rather than being designed independently (also in `design_fir_crossover.m`):

$$h_H[n] = \delta[n-D] - h_L[n] \quad \Rightarrow \quad H_L(e^{j\omega}) + H_H(e^{j\omega}) = e^{-j\omega D}, \;\; D = \frac{N-1}{2}$$

This "complement-by-construction" strategy is intentional: it turns reconstruction into a deterministic property (pure delay) rather than a best-effort outcome. The expected group delay D is therefore known a priori and can be used directly as a reference in both validation and figure interpretation.

IIR LR4 (efficient, unity-magnitude sum; SOS for stability).

The IIR crossover follows standard Linkwitz–Riley 4th-order practice by cascading two matched 2nd-order Butterworth sections per branch (designed in `design_iir_crossover_lr4.m`). To improve numerical robustness, the resulting 4th-order transfer functions are converted to second-order sections (SOS) (`tf2sos`) and applied using `sosfilt` (in `apply_crossover.m`). This choice is critical in practice: SOS realizations are significantly less sensitive to coefficient quantization and round-off than high-order direct-form implementations. In the discrete-time setting, the LR4 target is a summed response that is approximately unity in magnitude over the audio band, while its phase is generally frequency dependent (i.e., the sum behaves as an all-pass in phase). This distinction is handled explicitly in the reconstruction checks described below.

Band processing, per-band EQ, and exports

Given an input clip x[n], the processing stage applies both crossover realizations to produce low- and high-band signals $y_L[n]$ and $y_H[n]$, and their recombined sum $y_\Sigma[n] = y_L[n] + y_H[n]$ (implemented in `apply_crossover.m` and orchestrated by `process_one_clip.m`). Per-band equalization is applied as an independent gain stage per branch (configured in dB; see `cfg_default.m`), and is computed before recombination to reflect a realistic active loudspeaker DSP workflow (branch tuning prior to summation). In the baseline used for the paper, EQ is deliberately limited to branch gains to keep interpretation unambiguous: any spectral change with EQ enabled is attributable to controlled scaling, not additional poles/zeros from parametric sections.

For listening checks and external verification, `process_one_clip.m` exports WAV files for each implementation in a structured folder layout (FIR/IIR subfolders and separate files for low/high/sum, with and without EQ). Exports are protected against clipping by using a single global scaling factor computed from the peak across all generated signals in the batch. This is a deliberate reproducibility choice: it preserves relative levels across branches and across EQ off/on conditions, avoiding misleading comparisons caused by per-file normalization.

Verification logic (what is checked, and why it is the correct reference)

Verification combines frequency-domain and time-domain checks, with references chosen to match the theoretical behavior of each crossover family.

Frequency-domain checks: The project verifies branch magnitudes and the summed behavior. FIR validation targets a summed transfer equal to a **pure delay** $e^{-j\omega D}$ (not merely $|H_\Sigma| \approx 1$), while IIR LR4 validation targets a summed magnitude $|H_\Sigma| \approx 1$), over the audio band. These checks correspond directly to the plots reported in the Results section.

Time-domain reconstruction with consistent references: Reconstruction metrics are computed using integer-lag alignment based on cross-correlation (implemented in `recon_metrics.m`). This procedure is appropriate for the FIR sum because the expected difference is exactly an integer-sample delay. For the IIR LR4 sum, however, unity-magnitude does not imply a pure delay: the sum is generally an all-pass in phase. Therefore, comparing $y_\Sigma[n]$ directly to $x[n]$ can produce a non-zero "error" even when the implementation is correct. To avoid misinterpretation, the figure pipeline constructs an additional IIR reference by filtering $x[n]$ through the exact sum-path transfer function

$$H_\Sigma(z) = H_L(z) + H_H(z)$$

formed with a common-denominator representation (computed inside the figure script, i.e., within `make_all_figures.m`). The IIR reconstruction error is then evaluated against this sum-path reference, making the validation logically consistent: FIR is judged against a pure-delay target, while IIR is judged against its unity-magnitude sum-path equivalent.

Group delay plots with passband masking: Group delay is computed via `grpdelay` for the branches and for the summed response, but plotted only where it is meaningful. In deep stopband regions, where |H| is extremely small, phase (and hence its derivative) becomes numerically unstable and physically irrelevant. For this reason, the plotting pipeline masks group-delay values below a configurable magnitude threshold (default −40 dB), preventing artificial spikes and focusing the discussion on the frequency region that actually contributes energy to the output (implemented in the figure generation script).

One-command figure regeneration

All figures used in the paper are generated through a single entry-point script (`make_all_figures.m`), which executes design, processing, validation, and export in a deterministic order and stores results in the configured `figures` directory. This supports a strict paper workflow: any modification to the configuration (e.g., f_c, FIR length, EQ gains) can be propagated to every plot by rerunning one command, ensuring that captions and claims remain aligned with the actual implementation.

Validation criteria and interpretation of reconstruction behavior

Although both crossover implementations aim at splitting the signal into complementary frequency bands and recombining them into a controlled output, the correct interpretation of reconstruction behavior depends critically on the underlying theoretical target of each design. For this reason, different validation references are required for FIR and IIR crossovers, even when their summed magnitude responses appear equally flat.

For the complementary FIR crossover, the high-pass branch is explicitly constructed as the spectral complement of the low-pass branch, resulting in a summed transfer function equal to a pure delay. Consequently, the expected reconstructed signal is simply a delayed version of the input. In this case, time-domain validation can be performed by aligning the summed output with the input using an integer-sample delay and directly evaluating the residual error. A near-zero reconstruction error after alignment therefore confirms correct implementation and numerical consistency with the theoretical model.

In contrast, the IIR Linkwitz–Riley LR4 crossover is designed to guarantee unity magnitude of the summed response, but not linear phase. The sum of the low-pass and high-pass branches behaves as an all-pass system whose phase is generally frequency dependent. As a result, even a perfectly implemented LR4 crossover does not reconstruct the input signal as a pure delay. Direct comparison between the summed output and the original input (even after integer alignment) therefore produces a structured, non-zero difference that reflects phase reshaping rather than an implementation error.

To avoid misinterpretation, reconstruction validation for the IIR case must use a reference signal filtered through the exact summed transfer function of the crossover. When the summed output is compared against this all-pass reference, the reconstruction error correctly collapses to numerical precision, confirming that the implementation behaves as expected. This distinction is essential for correctly interpreting time-domain error plots and for separating true design limitations from plotting or reference artifacts. Group-delay analysis provides a complementary interpretation framework. Linear-phase FIR branches exhibit approximately constant group delay within their passbands, leading to a summed response with predictable delay behavior. In contrast, the IIR LR4 branches and their sum exhibit frequency-dependent group delay, consistent with their all-pass phase behavior. Group-delay values in deep stopband regions are not physically meaningful and are therefore masked in the reported figures to prevent misleading conclusions.

By explicitly distinguishing between magnitude-flat and delay-perfect reconstruction targets, the validation framework adopted in this work ensures that both FIR and IIR crossovers are evaluated against criteria that are theoretically correct and practically relevant. This allows a fair comparison between implementations and provides a clear basis for interpreting the results presented in the following section.

4. Results

The results are organized to mirror the validation logic: (i) frequency-domain verification of branch responses and summed behavior, (ii) time-domain reconstruction and interpretation of non-linear phase effects, and (iii) band-separated spectra and time signals with and without per-band EQ. All figures are generated by the unified pipeline described in Section 3 and saved automatically to the `figures` directory.

4.1 Frequency-domain crossover verification (baseline, EQ off)

FIR branch and sum magnitude (Fig 1).

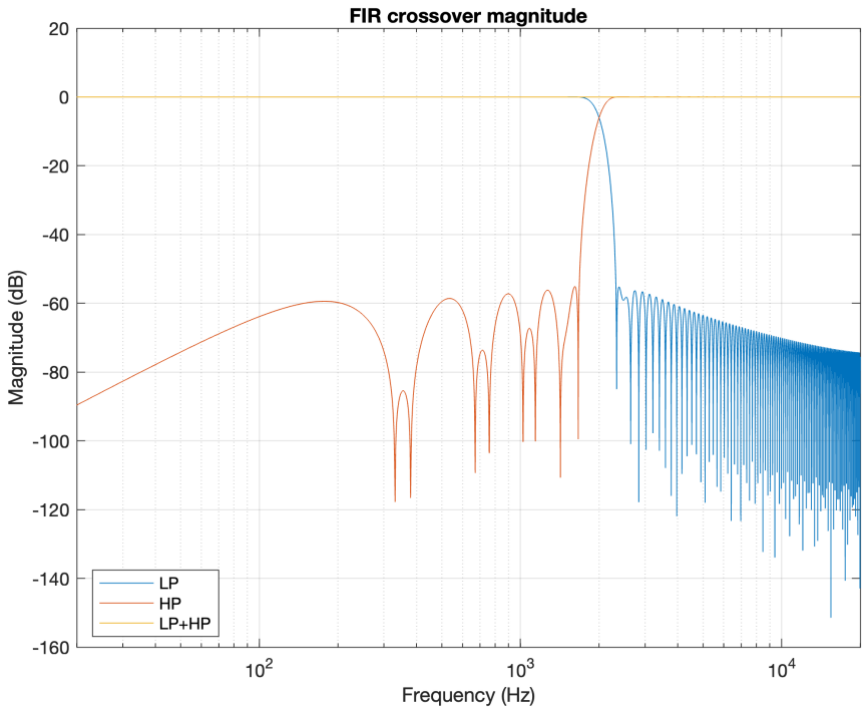


Figure 1 reports the FIR low-pass, high-pass, and summed magnitude responses. The key observation is not only that the two branches provide the intended split around the crossover region, but also that the summed response remains flat, consistent with a complementary construction. This confirms that the FIR crossover achieves the target "split–recombine" behavior in magnitude without requiring any additional tuning.

IIR LR4 branch and sum magnitude (Fig. 2).

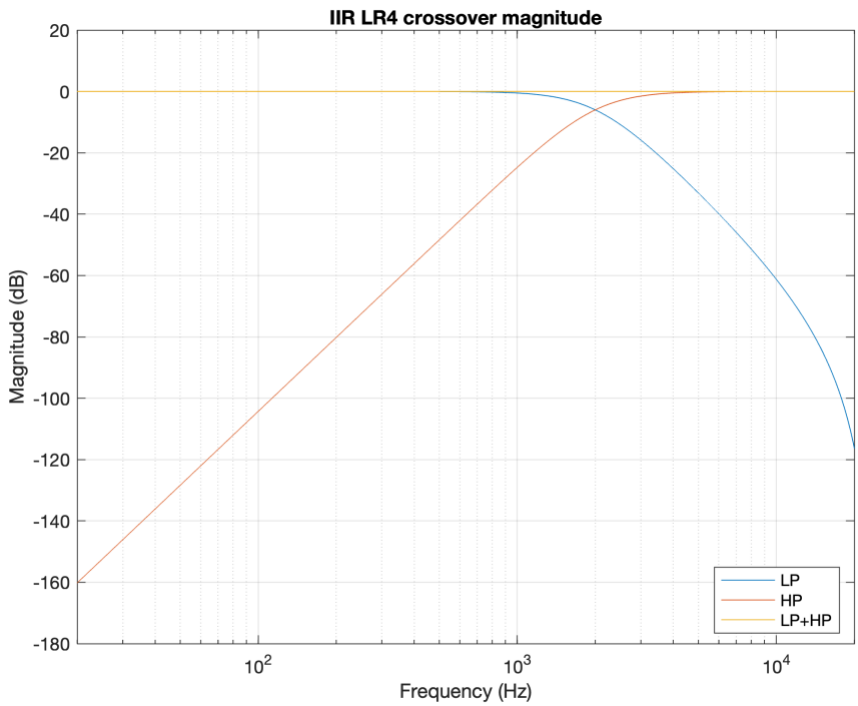
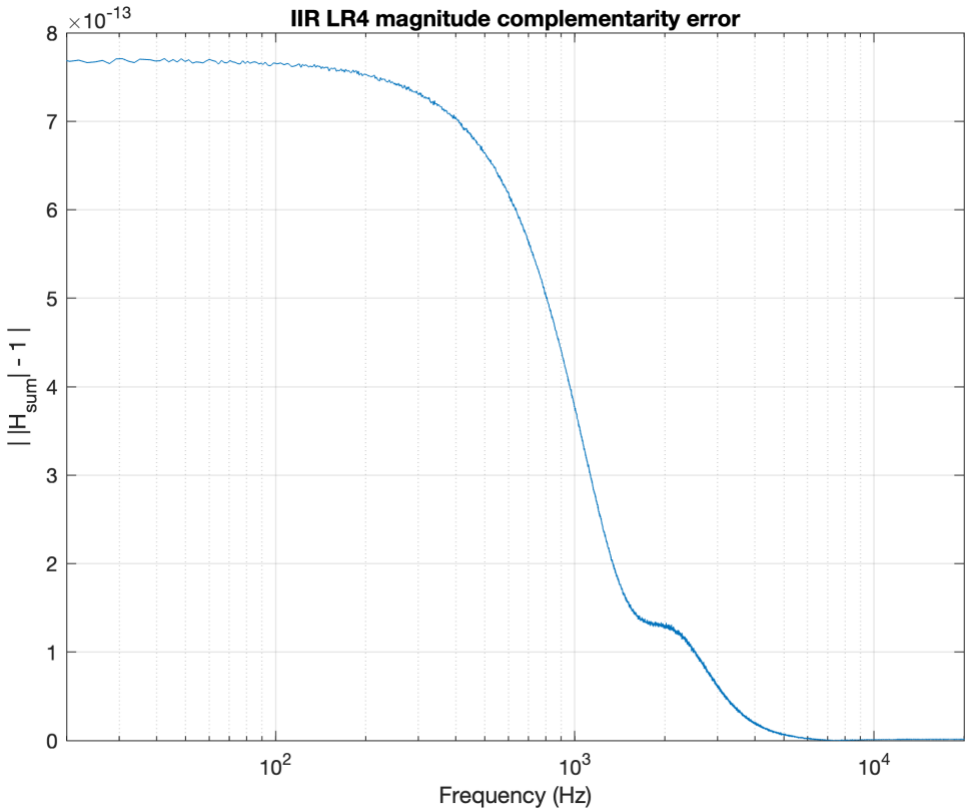


Figure 2 shows the corresponding responses for the IIR LR4 implementation. As expected for LR4, the low-pass and high-pass slopes are steeper than in the FIR baseline for comparable

transition width, and the summed magnitude remains essentially flat across the audio band. This figure establishes that both implementations satisfy the primary crossover requirement in the magnitude domain: band separation with controlled recombination.

Summed-magnitude deviation for IIR (Fig. 3).



It reports the deviation of the summed IIR response from unity magnitude, computed as

$$\left|H_{LP}\left(e^{j\omega}\right) + H_{HP}\left(e^{j\omega}\right)\right| - 1$$

over the audio band. The observed error remains at numerical precision, confirming that the Linkwitz–Riley LR4 implementation satisfies the intended magnitude-complementarity condition.

It is important to note that this verification is strictly limited to the magnitude domain: a unity-magnitude sum does not impose any constraint on the phase of the combined response. Consequently, this result alone does not imply time-domain reconstruction equivalent to a pure delay, but only guarantees that the spectral magnitude of the recombined signal is preserved..

4.2 Time-domain reconstruction and phase interpretation (baseline, EQ off)

Reconstruction error comparison (Fig. 4)

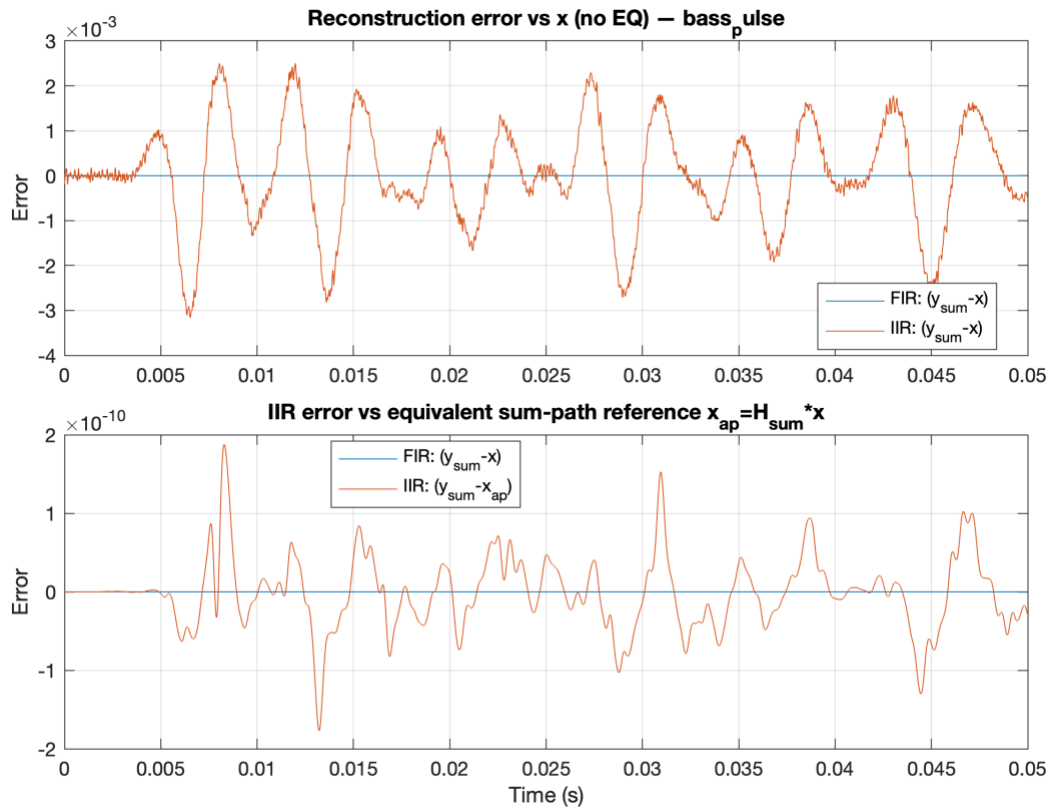


Figure 4 is the most diagnostic time-domain plot. The FIR reconstruction error is near zero after integer-sample alignment, which is consistent with the FIR complement-by-construction property (sum equals a pure delay). The IIR curve, instead, exhibits a structured non-zero difference if the comparison is performed directly against the unfiltered input: this does not indicate a design flaw in the LR4 magnitude response; it reflects the fact that the LR4 sum is typically unity in magnitude while its phase is frequency dependent (all-pass-like). In other words, the IIR sum preserves spectral magnitude but reshapes phase, so “matching the input up to a delay” is not the correct reference unless the sum-phase is itself a pure delay. This plot motivates the group-delay analysis and clarifies why magnitude-only checks (Fig. 2–3) are insufficient to predict time-domain identity.

Time-domain branch signals and sum (Fig. 5)

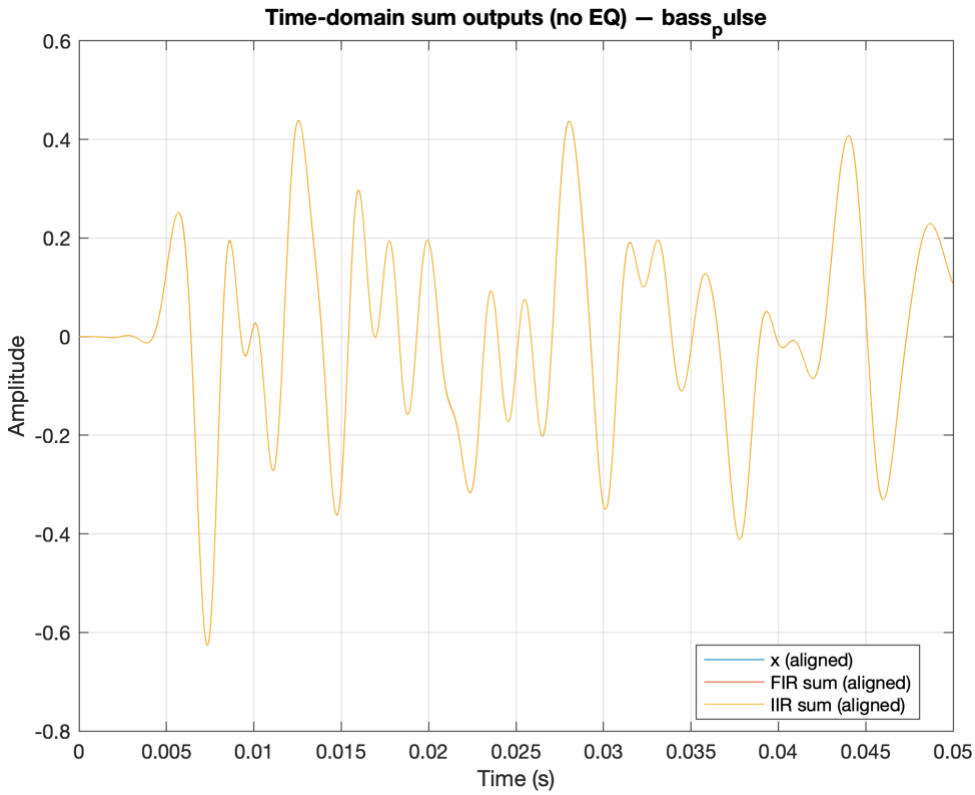


Figure 5 provides an intuitive view of the split in the time domain: the input is decomposed into low and high components, then recombined. This plot is not meant to prove complementarity by itself; instead, it helps build intuition for how energy is redistributed between branches and why the recombined sum can still match the original even when the individual branches appear very different.

4.3 Spectral decomposition after crossover (EQ off)

FIR: sum / low / high spectra (Fig. 6)

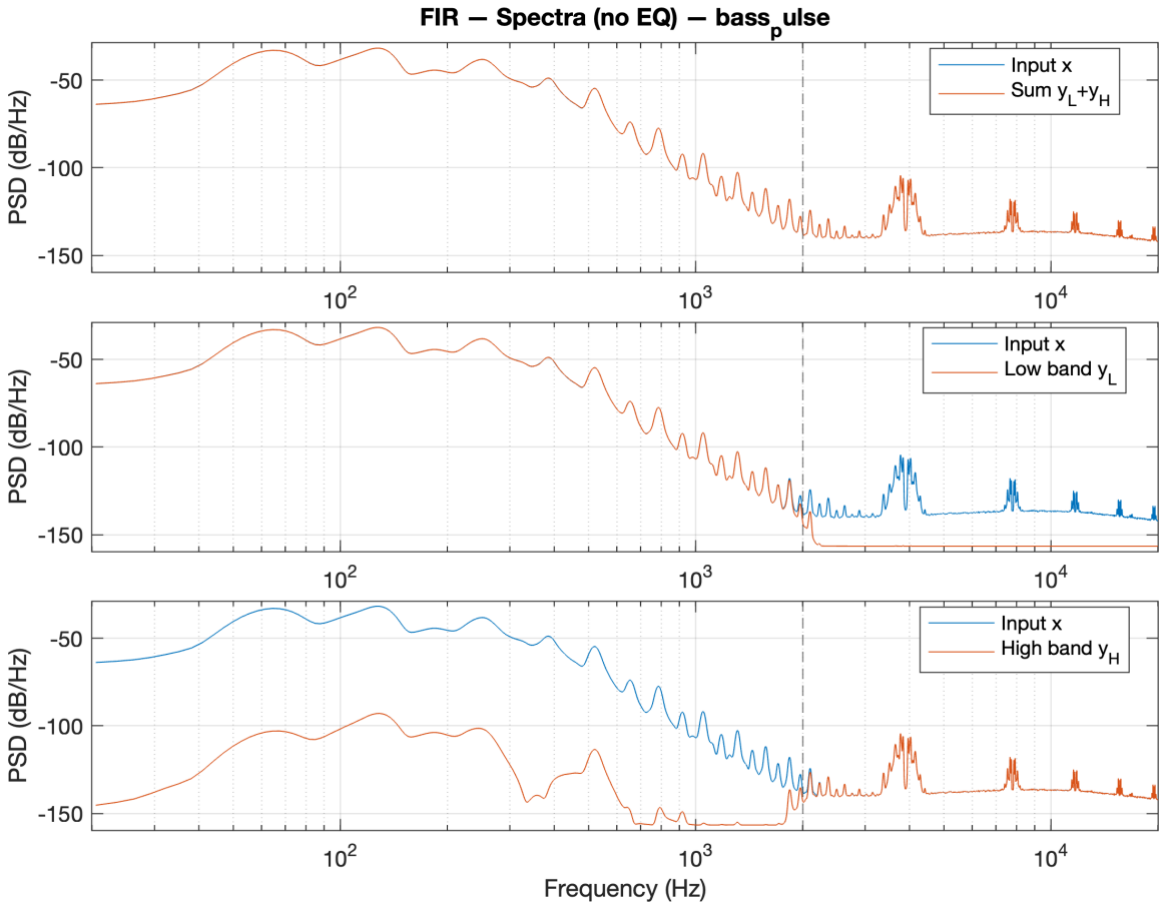


Figure 6 summarizes the FIR behavior on a representative clip using a three-panel comparison: summed output spectrum, low-band spectrum, and high-band spectrum. The low band concentrates energy below the crossover region while the high band carries primarily the upper-frequency content. The summed spectrum remains consistent with the original broadband content, confirming that the crossover is separating energy as intended without introducing visible magnitude anomalies in the recombined output.

IIR: sum / low / high spectra (Fig. 7)

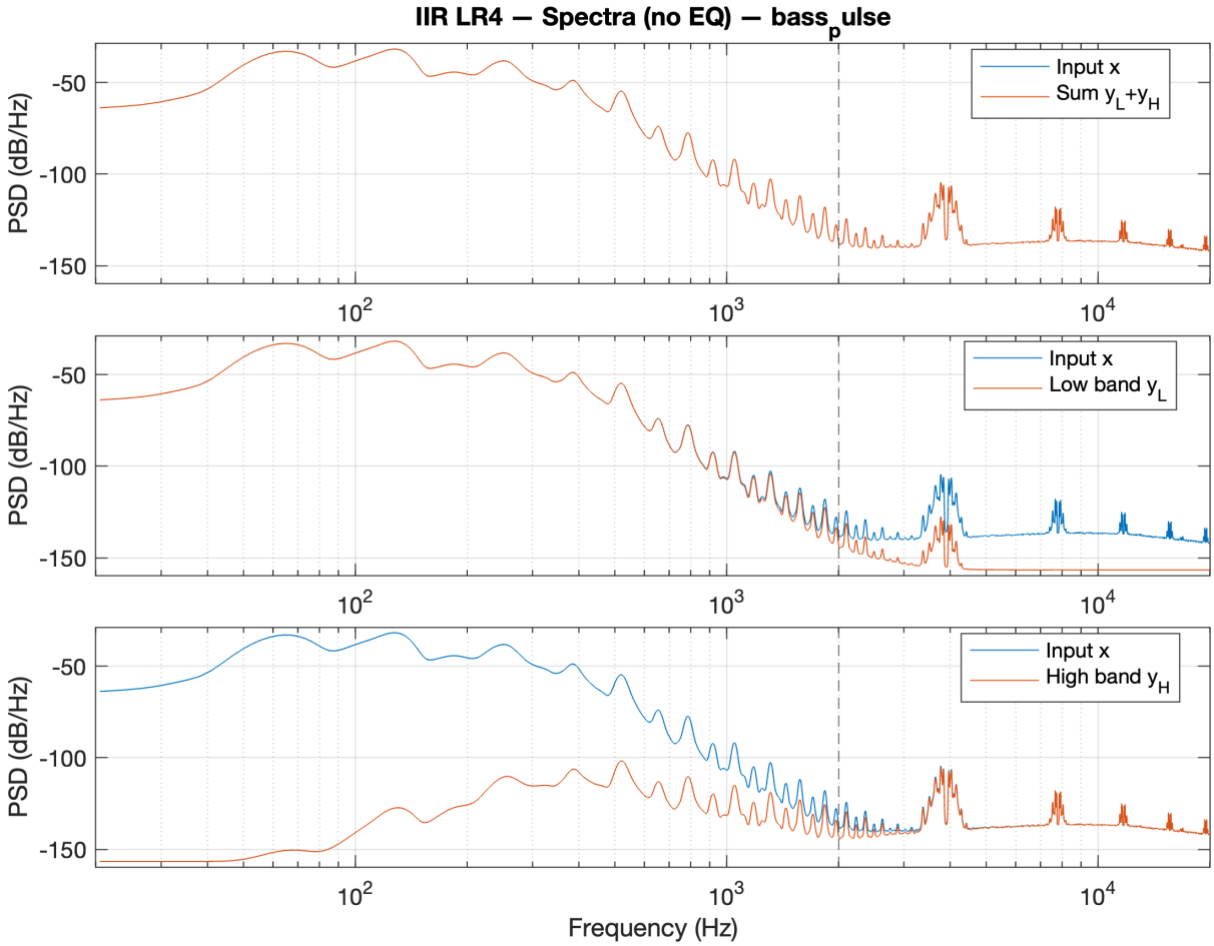


Figure 7 provides the same decomposition for the IIR LR4. The key role of this figure is comparative: it shows that, even though FIR and IIR can both achieve a flat summed magnitude response, the detailed distribution near the crossover region and the transition behavior can differ. This establishes an important practical point: selecting FIR vs IIR is not about "works vs does not work," but about trading off phase/latency/complexity while keeping magnitude behavior acceptable.

4.4 Per-band EQ enabled: spectral and time-domain effects

FIR with per-band EQ (Fig. 8)

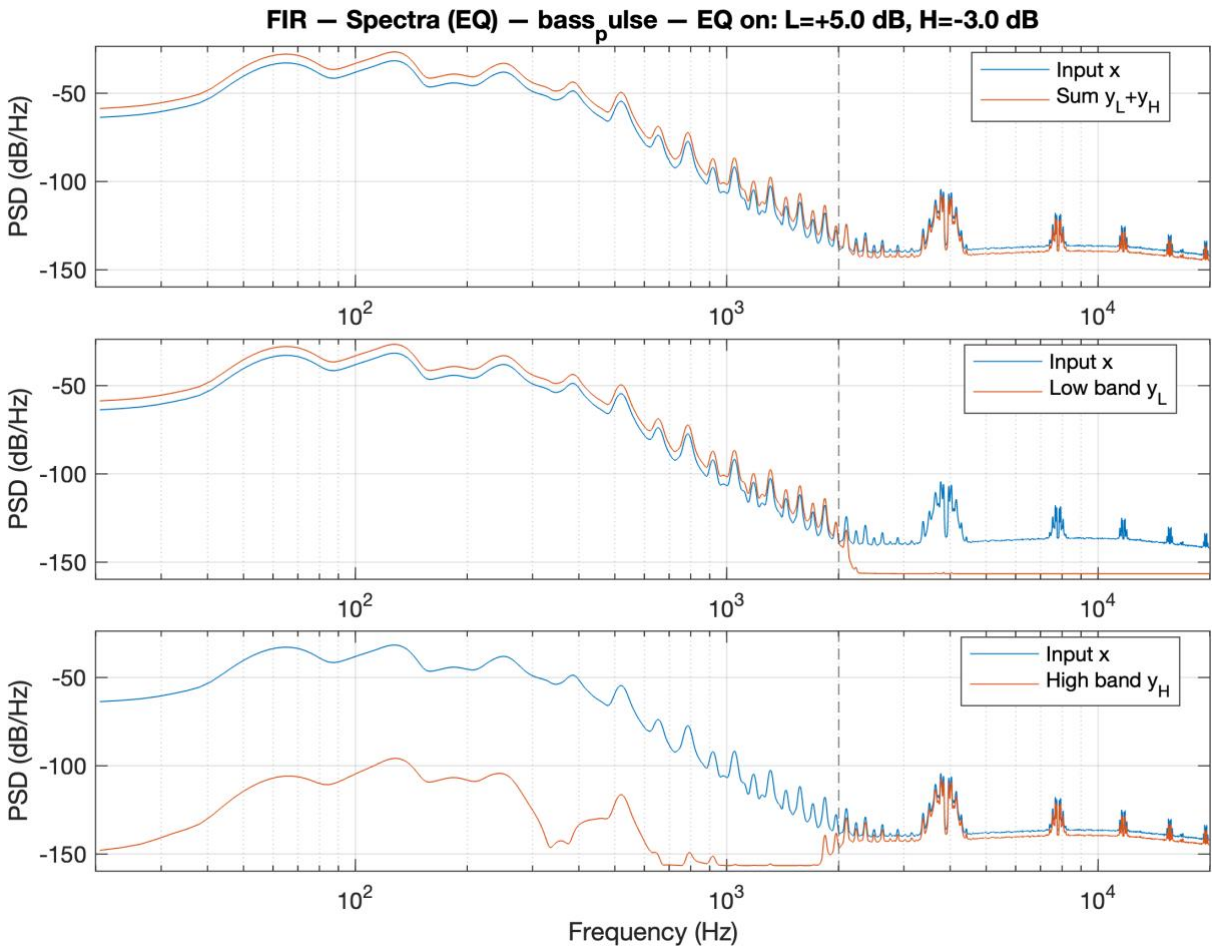


Figure 8 shows the FIR decomposition after enabling per-band EQ (implemented as independent branch gains in the baseline). The low and high spectra shift according to the applied gains, and the summed spectrum changes accordingly. This is expected and desirable: the purpose of per-band EQ in active systems is to shape the final response by adjusting branches independently. The figure demonstrates that EQ acts "where intended" (within branches) without breaking the crossover split structure.

IIR with per-band EQ (Fig. 9)

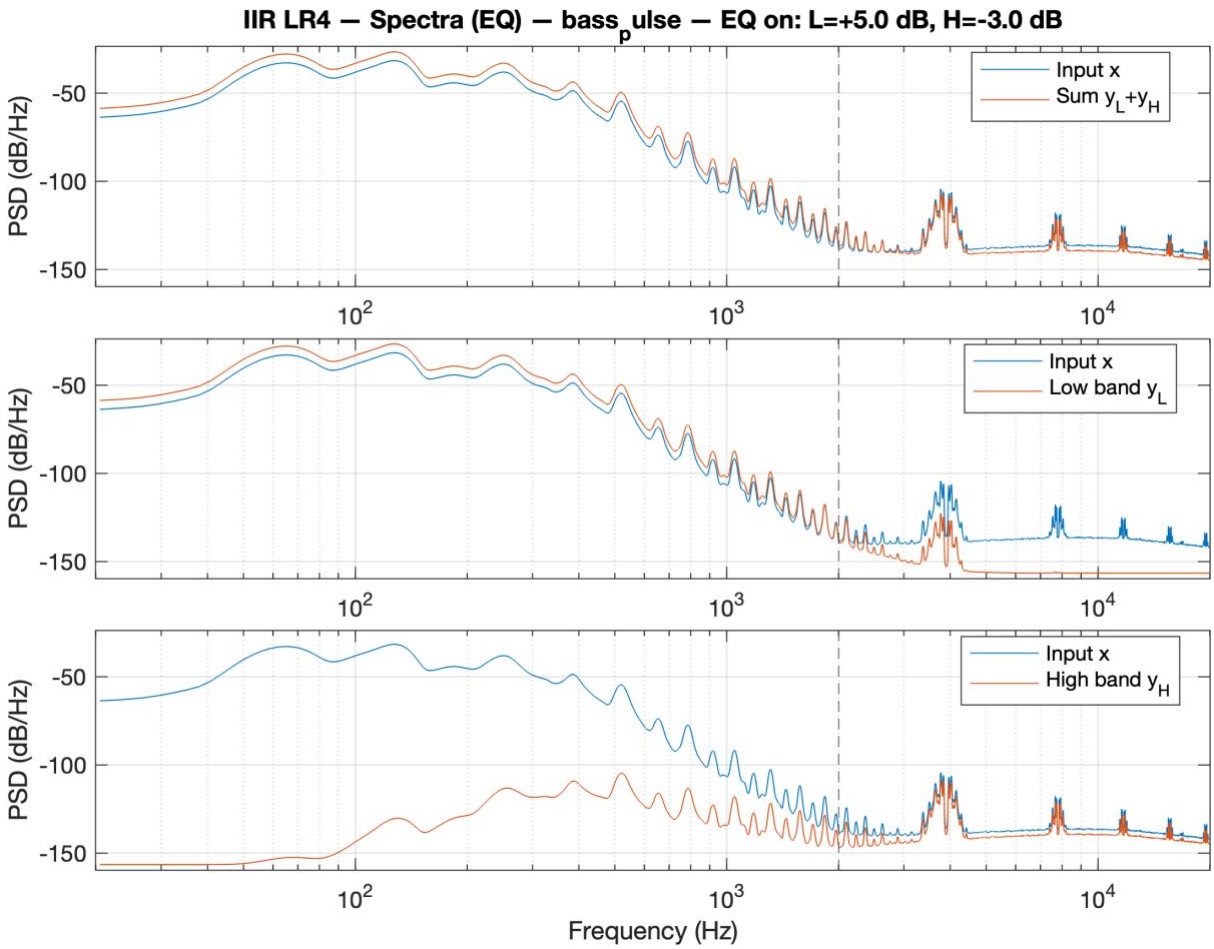
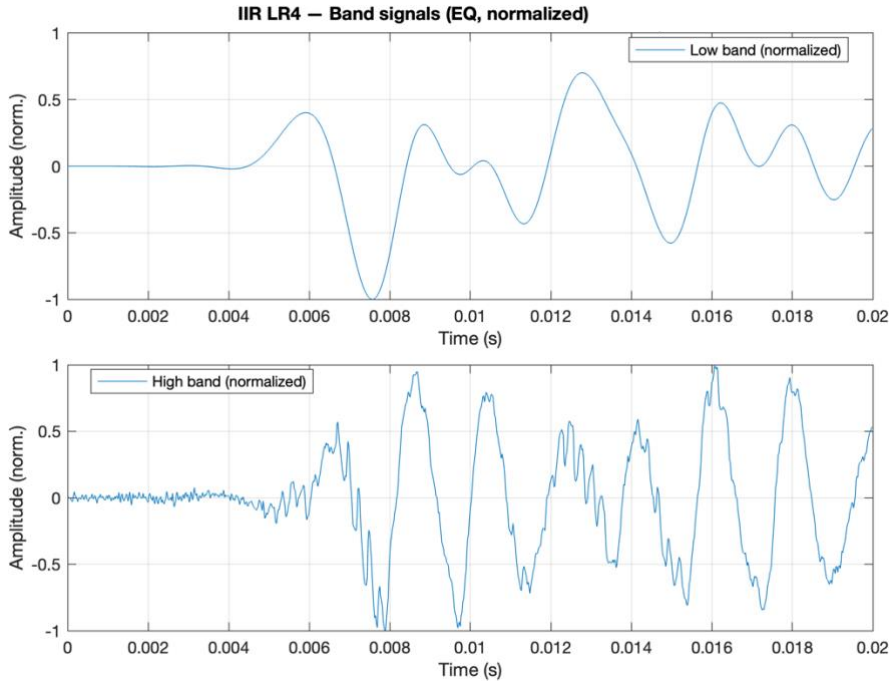
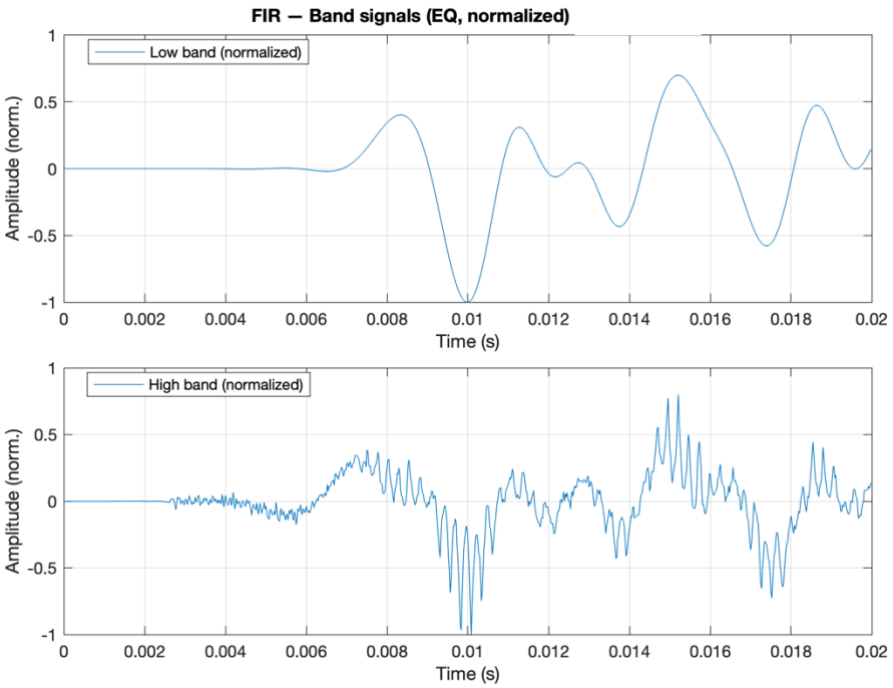


Figure 9 repeats the EQ case for IIR LR4. The interpretation is analogous: EQ modifies branch levels and therefore modifies the summed spectrum. The relevance of this figure is that it confirms the EQ stage is implementation-agnostic: both FIR and IIR pipelines support the same tuning workflow (split → tune per band → sum), allowing fair comparisons under realistic conditions.

Time-domain branch signals with EQ (Fig. 10–11)





Figures 10 and 11 show the low- and high-band signals in the time domain after per-band equalization, for the IIR and FIR implementations respectively. These plots are intended as qualitative visualizations of band isolation and relative level after tuning, rather than as direct indicators of signal quality. For audio material dominated by low-frequency content, the high-band waveform may appear highly irregular or noise-like; this effect is primarily due to the much lower absolute energy of the high band, combined with transient leakage and visualization scaling. To improve readability, waveforms are normalized per band for display

purposes only. The correct interpretation of band behavior and EQ impact is therefore provided by the corresponding spectral decomposition figures, which show that energy remains confined to the intended frequency regions.

4.5 Group-delay interpretation (branch and sum)

Branch group delays (Fig. 12)

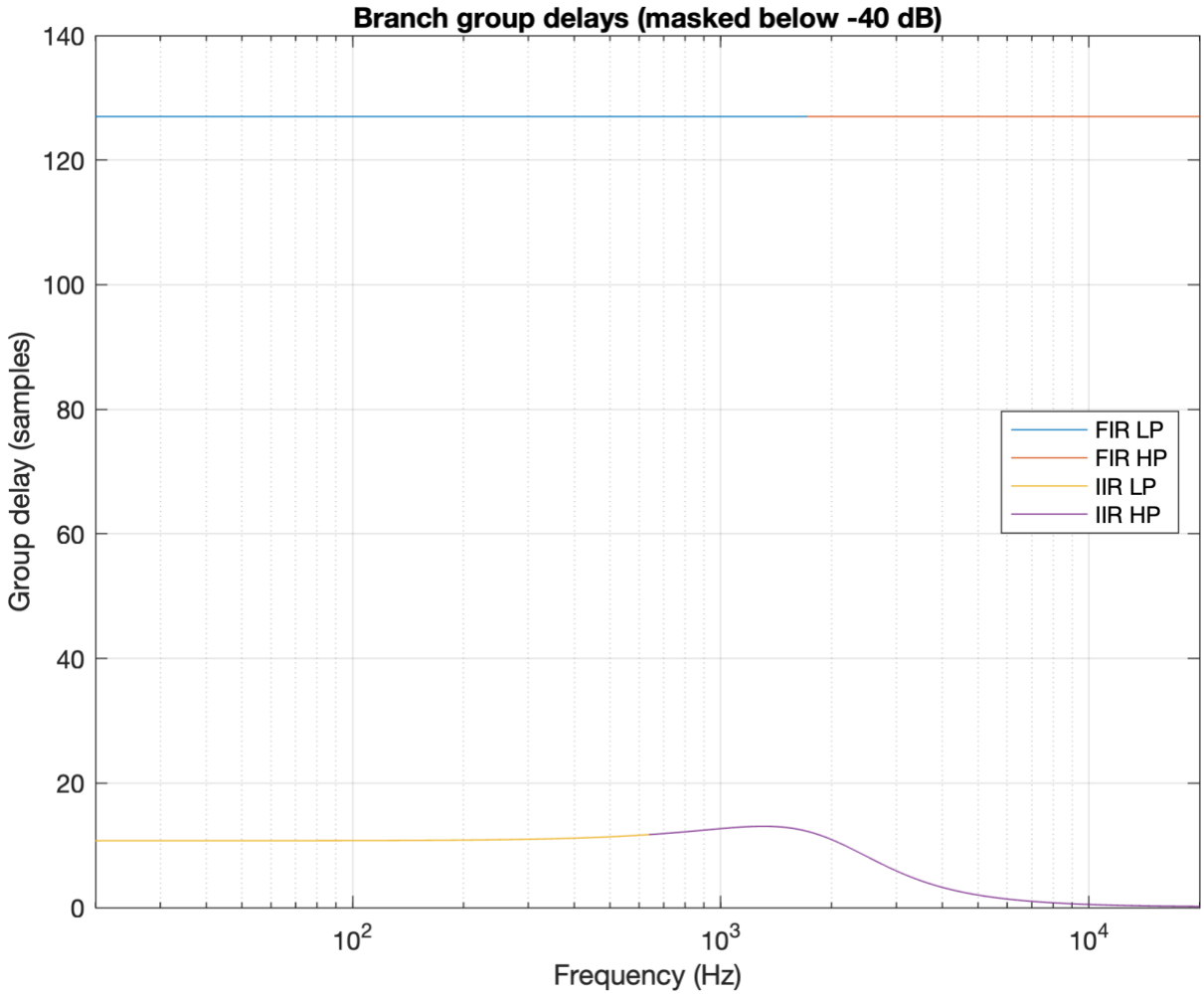


Figure 12 compares the frequency-dependent delay behavior of the low and high branches. The plot is masked in stopband regions (where group delay is not meaningful) to avoid numerical artifacts. The qualitative result is that the FIR branches exhibit near-constant delay in their passbands (consistent with linear-phase behavior), while the IIR branches show frequency-dependent delay. This is the time-domain counterpart of the “efficiency vs linear-phase” trade-off: IIR achieves low computational cost and low latency, at the expense of branch phase linearity.

Summed-response group delay (Fig. 13)

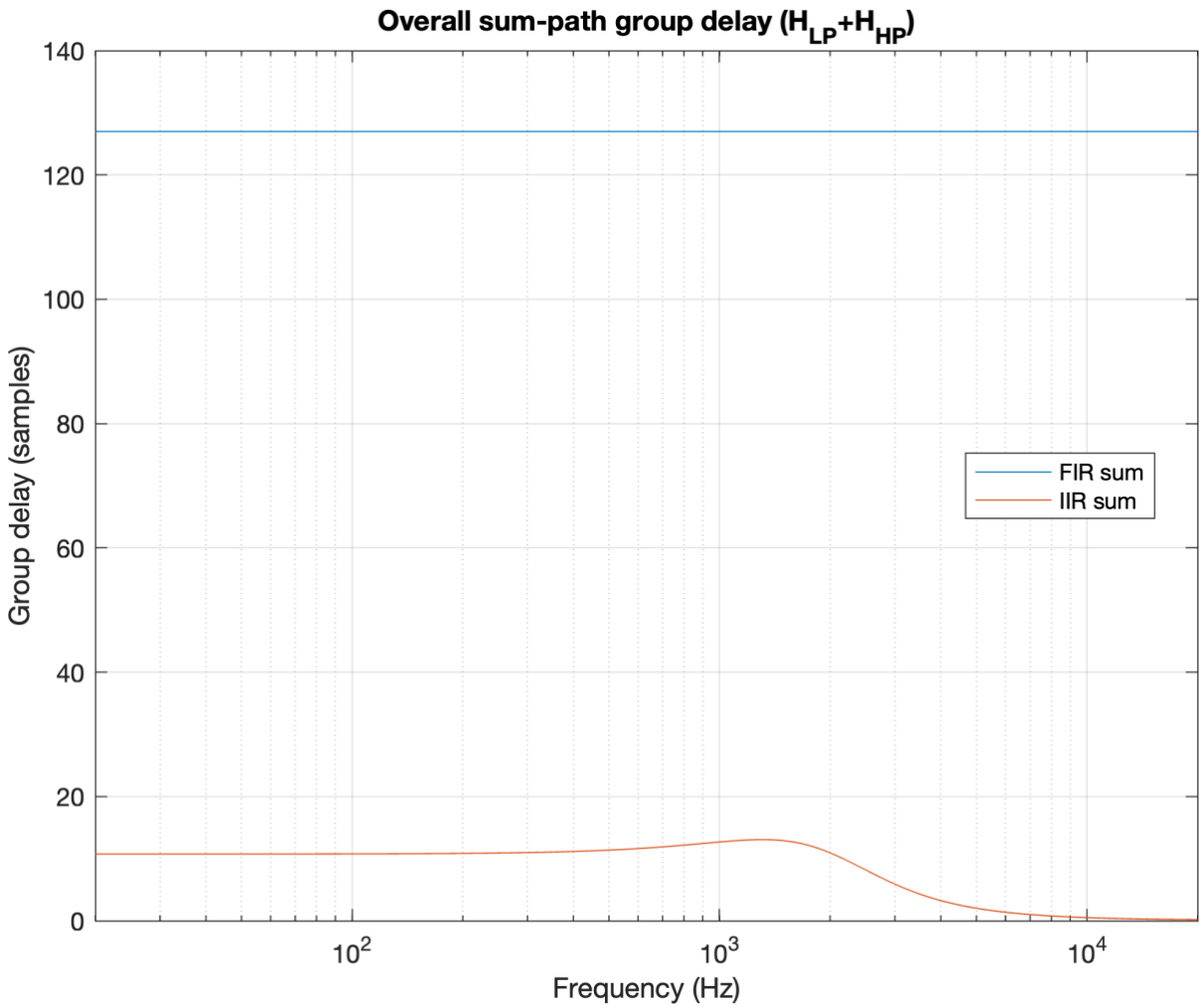


Figure 13 focuses on the summed path. For the FIR case, the group delay is approximately constant, matching the "pure delay" reconstruction behavior observed in Fig. 4. For the IIR case, the summed group delay is frequency dependent, which explains why a unity-magnitude sum can still differ from the input in the time domain. This figure provides the physical interpretation that connects frequency-domain checks (Fig. 2–3) to reconstruction-error behavior (Fig. 4).

4.6 Takeaways from the plots (selection criteria FIR vs IIR)

The figure set enables a practical selection of FIR versus IIR crossovers based on measurable behaviors rather than general preferences.

Magnitude integrity is necessary but not sufficient. Figures 1–3 confirm that both approaches can achieve the core magnitude-domain requirement: a controlled split and a flat summed magnitude. However, Figure 4 shows that this does not imply time-domain identity. Therefore, a magnitude-only validation (flat $|H_\Sigma|$) should be treated as a baseline check, not a final acceptance criterion.

Time-domain fidelity depends on the sum-phase behavior. The FIR implementation is delay-perfect by construction: after integer-sample alignment, the sum matches the input with negligible residual (Fig. 4), and the summed group delay is approximately constant (Fig. 13). This makes FIR preferable when the downstream system benefits from predictable delay (e.g., multichannel alignment, strict transient preservation, or when consistent phase is a requirement). In contrast, the IIR LR4 sum is typically unity-magnitude with frequency-dependent phase (all-pass-like), which explains the structured time-domain mismatch when compared directly to the unfiltered input (Fig. 4) and the non-constant summed group delay (Fig. 13). This is not a defect but an intrinsic property of the chosen IIR topology.

Branch delay behavior informs integration constraints. The branch group-delay plot (Fig. 12, passband-masked) highlights a key practical difference: FIR branches present near-constant delay in their passbands, while IIR branches exhibit frequency-dependent delay. In systems where branch-to-branch temporal alignment matters (e.g., crossover-region blending, driver time alignment, or tight phase targets), this difference can influence tuning effort and achievable results.

Per-band EQ is validated as a workflow, not as a "flat-sum" requirement. Figures 6–11 demonstrate that the pipeline supports a realistic active-speaker procedure: split → tune branches → sum. With EQ enabled, the summed spectrum changes by design (Fig. 8–9), and the band time signals reflect the clip's content distribution (Fig. 10–11). For low-frequency-dominant material, the high-band waveform can appear visually "noisy" simply because its absolute amplitude is much smaller; the correct interpretation is given by the spectral triptychs rather than time-domain amplitude alone.

Practical decision rule derived from the plots.

- Choose **FIR** when constant delay / linear-phase behavior is a priority and the system can tolerate higher latency and computational cost. Use Fig. 4 and Fig. 13 as acceptance checks (near-zero reconstruction error after alignment; approximately constant summed group delay).
- Choose **IIR LR4** when low latency and efficiency are primary constraints and phase nonlinearity is acceptable. Use Fig. 2–3 to confirm unity-magnitude summation and Fig. 13 to document the expected frequency-dependent delay; interpret Fig. 4 with the correct all-pass reference rather than direct comparison to the raw input.

5. Conclusion

This work presented a compact and reproducible two-way digital crossover workflow for multi-driver audio systems, including an explicit per-band equalization stage and a controlled comparison between complementary FIR and IIR Linkwitz–Riley (LR4) implementations. The objective was not to propose a novel crossover topology, but to clarify through consistent validation criteria and representative plots how standard FIR and IIR solutions behave in both frequency and time domains under realistic tuning conditions.

Both implementations were shown to satisfy the primary magnitude-domain requirement of a crossover: correct band separation and controlled recombination. The complementary FIR design achieves this through exact spectral complementarity, resulting in a summed response equal to a pure delay. The IIR LR4 implementation, instead, achieves a unity-magnitude sum

over the audio band, consistent with its widespread use in active loudspeaker systems. Frequency-domain checks alone therefore confirm that both approaches are valid from a magnitude perspective.

The central distinction between the two designs emerges in the time-domain interpretation. The FIR crossover provides delay-perfect reconstruction by construction, with approximately constant group delay across the passband. In contrast, the IIR LR4 crossover produces a summed response with frequency-dependent phase and group delay, behaving as an all-pass system. As a consequence, a flat summed magnitude does not imply time-domain identity with the input signal. Correct validation of the IIR case requires comparison against the appropriate all-pass reference rather than against a delayed input. This distinction is essential to avoid misinterpreting intrinsic phase behavior as an implementation error.

The per-band equalization stage was verified as an effective and implementation-agnostic tuning mechanism. Independent gain adjustments applied to the crossover branches reshape the summed spectrum as intended, without breaking the split–recombine structure. Spectral decompositions confirm that energy remains confined to the expected frequency regions, while time-domain band signals reflect the relative energy content of the source material rather than distortion artifacts.

The results lead to a clear selection criterion. Complementary FIR crossovers are preferable when predictable delay, linear-phase behavior, or strict time-domain alignment are required, at the cost of higher latency and computational complexity. IIR LR4 crossovers are preferable when low latency and efficiency dominate the design constraints, and frequency-dependent phase behavior is acceptable or can be accommodated at the system level. The validation framework and figure set presented in this work provide a practical basis for making this choice in real-world active loudspeaker and audio DSP applications.

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